



PULSE

Profit Pool Intelligence Engine

A Bayesian simulation platform that transforms
strategic force assessment into decision intelligence.

Where will your **profit pools** be in five years?

The companies that answer this question first
will define the next decade of consumer goods.

Profit pools are shifting — fast

Regulation

EU Green Deal, PFAS bans, EPR mandates
are rewriting category economics

Technology

AI, biotech, DTC platforms disrupt
every step of the value chain

Sustainability

Climate costs, water scarcity, packaging
levies hit the bottom line

Consumer

Clean beauty, premiumization, value
polarization reshape demand curves

Retail Power

Private label growth and channel
consolidation squeeze branded margins

Competition

Challenger brands, cross-category entry,
and M&A redraw the competitive map

Most companies react. Leaders anticipate.

Typical Approach

- Static, annual assessments
- Point estimates without uncertainty
- Forces analyzed in isolation
- No competitive response modeling
- Backward-looking calibration gap

What's Needed

- ✓ Living, probabilistic simulation
- ✓ Full uncertainty quantification
- ✓ Causal force propagation
- ✓ Game-theoretic competitor modeling
- ✓ Backtested for credibility

PULSE

A privacy-first, AI-augmented simulation engine that models how strategic forces reshape profit pools across categories and time.

01 Simulate

Bayesian Monte Carlo
with 10,000+ scenarios

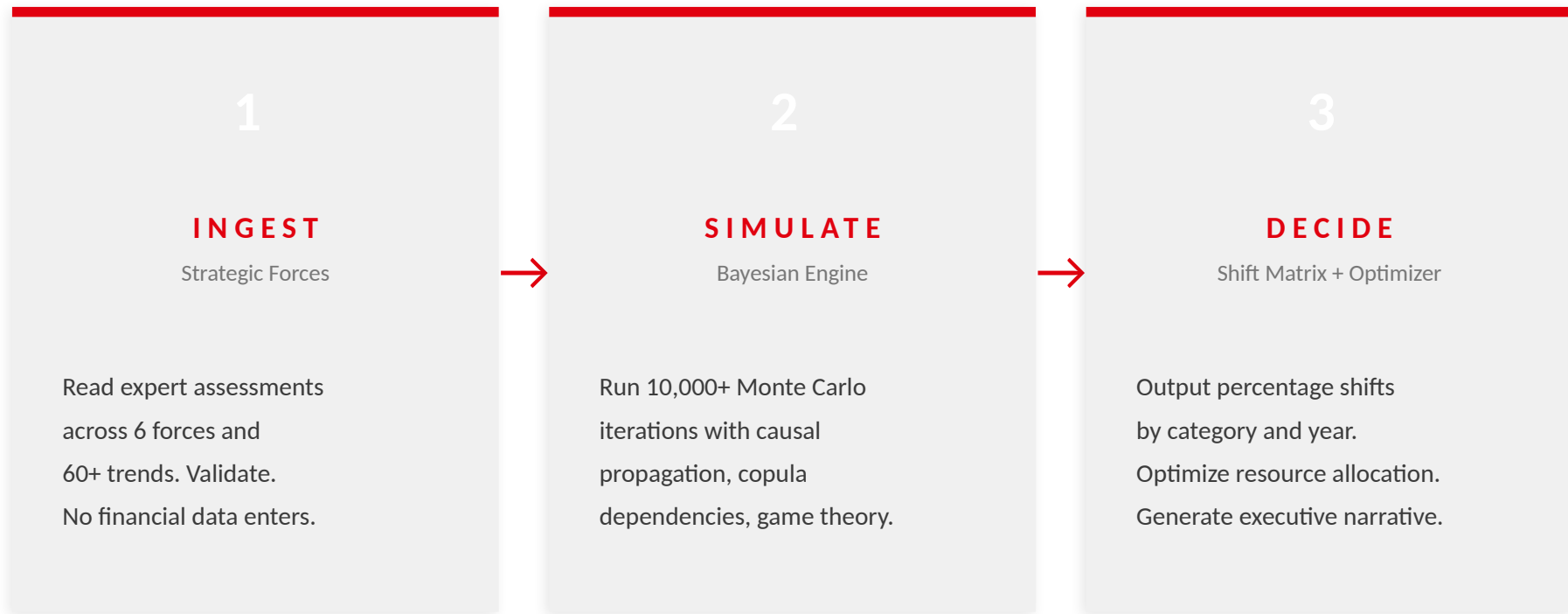
02 Connect

Causal DAG maps how
forces propagate

03 Decide

Resource allocation
optimizer for action

How PULSE works



Six forces shape every profit pool

Consumer

Demand shifts, premiumization, clean beauty, health consciousness

Customer

Retail consolidation, private label growth, channel power dynamics

Technology

AI/biotech, DTC platforms, smart manufacturing, digital commerce

Government

EU Green Deal, PFAS regulation, EPR mandates, compliance costs

Environmental

Climate impact, water scarcity, packaging levies, sustainability pressure

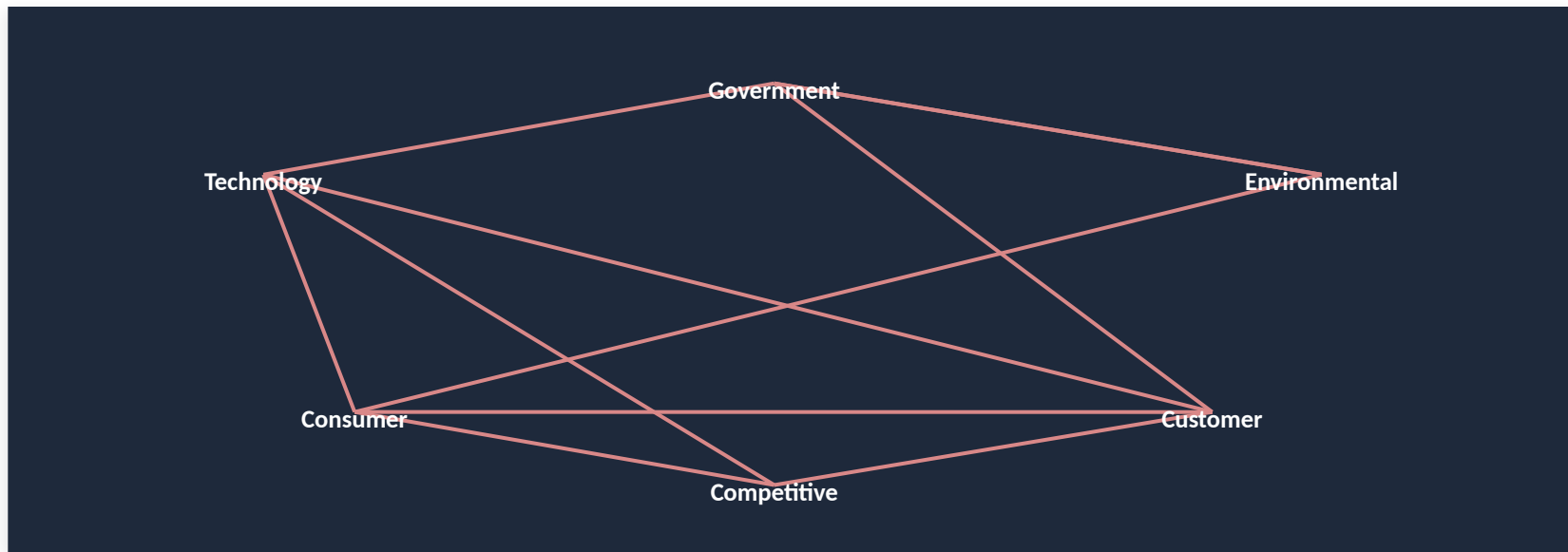
Competitive

Challenger brands, M&A activity, cross-category entry, price wars

Each force contains 8–12 individually scored trends — 60+ in total across all categories.

Forces don't act alone — they propagate

PULSE uses a Causal DAG to model how a shock in one force propagates to others — with calibrated weights and time lags.



Example: EU regulation (Gov) → reformulation costs (Tech) → shelf price (Customer) → demand shift (Consumer)

Not guesses — Bayesian probabilities

Traditional approaches give you a single number.

PULSE gives you a full probability distribution.

Bayesian Priors

Start with expert knowledge, update with evidence

Copula Dependencies

Model how risks cluster together in crises

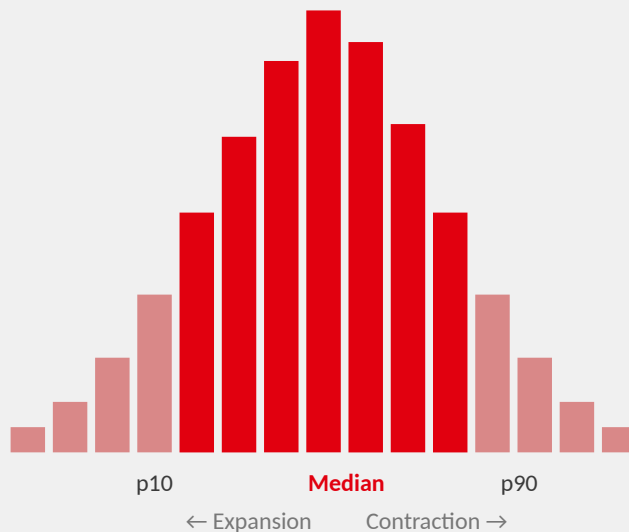
10,000+ Simulations

Full posterior distribution, not a point estimate

Convergence Tested

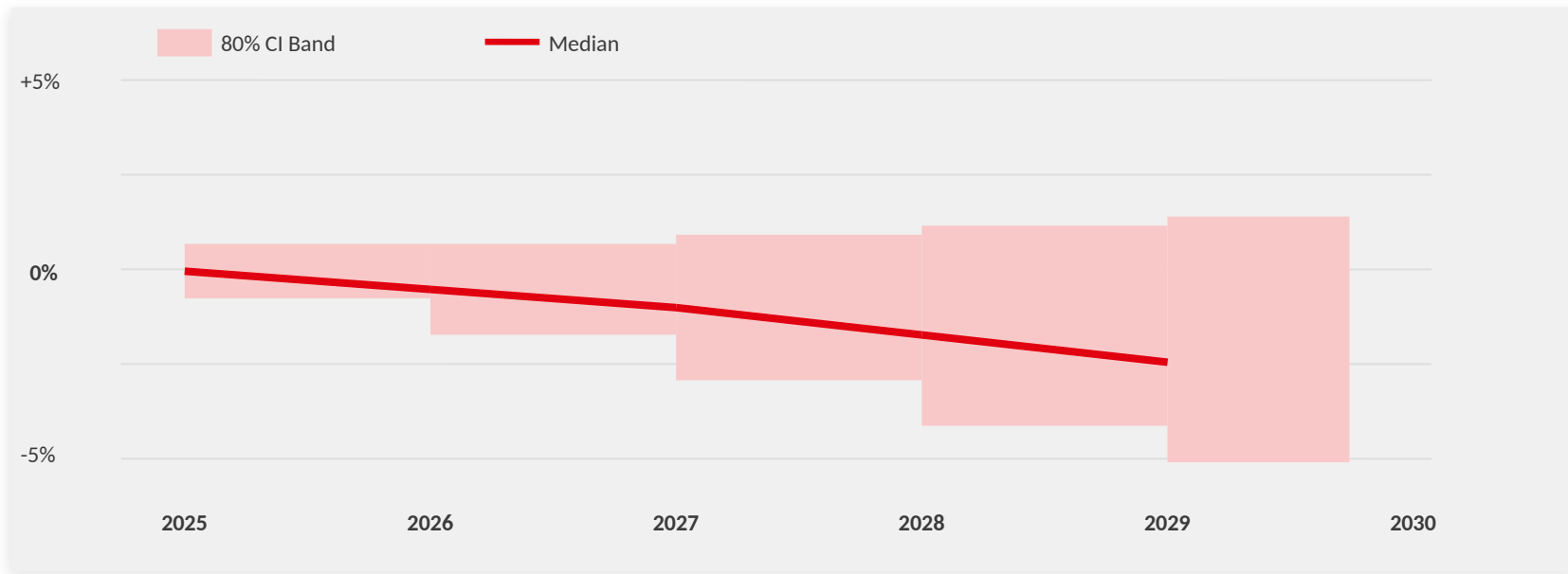
Gelman-Rubin R -hat < 1.05 for statistical rigor

Posterior Distribution



Five-year paths, not two-point snapshots

PULSE models continuous trajectories from 2026 to 2030. Velocity tracking and early-warning triggers enable path-dependent strategy.



One output: the Shift Matrix

A single table of percentage shifts by category, year, and confidence level. Apply directly to financial models.

Category	2026	2027	2028	2029	2030
Hair: Color	-0.3%	-0.8%	-1.4%	-2.3%	-3.2%
Hair: Care	+0.2%	+0.5%	+1.1%	+1.8%	+2.6%
LHC: FCN	-0.5%	-1.2%	-2.1%	-3.5%	-4.8%
LHC: ADW	+0.1%	+0.4%	+0.9%	+1.5%	+2.1%
LHC: IC	+0.0%	+0.1%	+0.2%	+0.3%	+0.4%

Values shown are median shifts. Full output includes p10, p25, p75, p90 bands plus causal decomposition.

Application: $GP1_{\text{projected}} = GP1_{\text{actual}} \times (1 + \text{shift } \%)$

Delphi Process: structured expert scoring

Replacing ad-hoc scoring with a rigorous, debiased, multi-round elicitation protocol.

Pre-Workshop

Calibration

Present known-outcome historical trends to each scorer. Detect systematic biases (optimism, anchoring, recency). Compute calibration factors.

Round 1

Independent Scoring

Each expert scores impact (1-5) and probability (1-5) for every trend — blind, without seeing others' scores. Documented rationale required.

Round 2

Calibrated Re-Scoring

Share anonymized score distribution across all scorers. Experts re-score with group context. Measure inter-rater reliability (Krippendorff's α).

Round 3

Consensus & Debiasing

Discuss outlier scores in structured session. Apply bias correction factors. Final score = weighted median by calibration accuracy.

Quality Gate: $\alpha > 0.67$ required | Bias-corrected | Full audit trail | Scores feed Bayesian priors

Competitors will respond — we model that

A game-theoretic layer models likely competitive responses. Iterative Nash equilibrium converges to a stable outcome.

Premium Defender

- Innovation-led defense
- Premiumize further
- Lobby & comply early

Sustainability Leader

- Portfolio rationalization
- Lead compliance
- Acquire disruptors

Category Specialist

- Defend core categories
- Selective tech adoption
- Minimum compliance

All inputs are public market intelligence — no competitor financials enter the model.

From analysis to action: allocation

Every output answers: so what should we do?

INVEST MORE

Hair: Care
LHC: ADW

Growing pools
with momentum

DEFEND

LHC: FCN
Hair: Color

Contracting pools
requiring protection

HARVEST

LHC: IC

Stable pools
with low growth

All recommendations are in relative weights — never absolute currency amounts.

The model earns credibility first

PULSE uses historical versions (V1-V11) and actual market data to calibrate parameters. It proves accuracy before predicting.

- Attenuation factor (derived, not assumed)
- Distribution shapes (learned from errors)
- Causal DAG weights (fitted to history)
- Correlation structure (empirical, not flat)
- Overall accuracy score (reported to ExCo)

Backtesting Report

73%

of historical shifts predicted
within the 80% confidence interval

Based on V1-V11 historical versions
vs. actual Euromonitor market data

Security by architecture, not by policy

ENTERS PULSE

- ✓ Trend names & descriptions
- ✓ Impact / probability scores (1-5)
- ✓ Direction (expansion / contraction)
- ✓ Category names (public knowledge)
- ✓ Causal propagation weights
- ✓ Public market data (Euromonitor)

NEVER ENTERS PULSE

- ✗ Net External Sales (NES)
- ✗ Gross Profit (GP1, GP2)
- ✗ Regional financial splits
- ✗ Cost structures (COGS, OpEx)
- ✗ Company market shares
- ✗ Management unit financials

Financials are structurally excluded — enforced by schema, firewall, and automated tests.

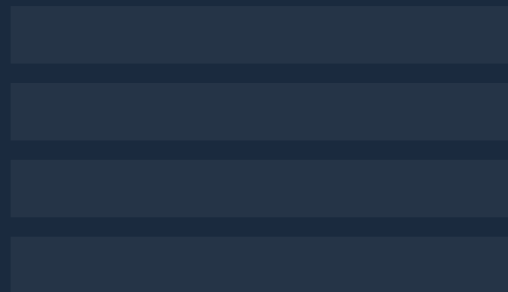
The War Room: one screen tells the story

Net Pool Shift: -3.1% | 80% CI: [-1.5%, -5.5%] | Accuracy: 73% | Velocity: Accelerating

Force × Category Heatmap



Contextual Panel



Path Timeline 2026 — 2030



Scenario engine: shock, propagate, decide

Base Case	Current expert scores, causal DAG active, no external shocks
Green Squeeze	Environmental shock → Government, Technology, Customer, Consumer
Tech Disruption	Technology shock → Consumer adoption, Competitive gaps, Channel shift
Price War	Competitive shock → Customer margin pressure, Consumer trading down
Regulatory Cascade	Government shock → propagates through DAG with lag structure
Perfect Storm	Correlated tail event — t-copula at 1st percentile (crisis scenario)

Custom scenarios: user shocks a specific force, the Causal DAG propagates the rest automatically.

AI augments judgment — never replaces it

Trend Scanner

Monitors 19 free-tier data sources continuously.
Flags new developments. Weekly digest.

GDELT, GNews, RSS, ECHA, EPO, Google Trends

Score Calibrator

Cross-validates expert scores for consistency.
Benchmarks against real-world trajectory data.

Internal consistency + external benchmarks

Scenario Narrator

Converts Shift Matrix into executive narrative.
Explains causal propagation in plain language.

LLM-generated from relative data only

Chat Interface

Natural language queries about the model.
Financial queries get redirected.

Conversational AI layer

Provider-agnostic: Claude API (MVP) → Azure OpenAI (production) via single config change.

Built on proven, enterprise-grade tech

Simulation Engine	Python 3.11+, NumPy, SciPy, Pandas, NetworkX	Bayesian Monte Carlo, copula dependencies, causal propagation
API & Data	FastAPI, Pydantic, SQLite, OpenPyXL	REST API, validation, zero-financial schema, audit trail
War Room UI	React 18, Vite, D3, Recharts, Tailwind, Framer Motion	Single-page War Room, 60fps animations, Apple-grade UX
AI Layer	LLM Provider ABC, Claude / Azure OpenAI	Trend scanning, score calibration,
Total running cost: €0/year — All 19 external APIs are free tier. No vendor lock-in.		

Building with DX: MVP to enterprise

MVP

Strategy Team

Working prototype:
Bayesian engine, CLI,
Shift Matrix output.
Proves the concept.



DX BUILD

Joint Team

Cloud deployment,
War Room dashboard,
API integration,
Security hardening.



ENTERPRISE

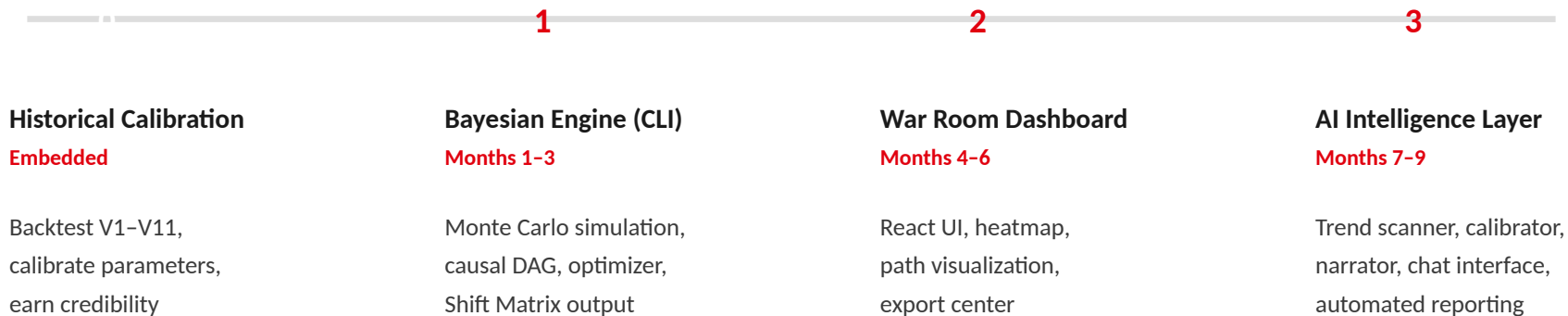
DX Owned

Azure tenant hosting,
SSO + access control,
Automated pipelines,
ExCo reporting.

DX Engagement Steps

- 1 Handoff: MVP code + spec to DX
- 2 Azure tenant setup + CI/CD pipeline
- 3 Security review + InfoSec approval
- 4 Dashboard deployment + SSO
- 5 Financial system integration
- 6 Ongoing: quarterly model updates

Phased delivery — standalone value at each step



Each phase delivers standalone value. Phase 1 alone provides a probabilistic, causally-structured simulation. Every subsequent phase compounds the advantage.

The Ask

- 1 Endorse PULSE as the strategic profit pool simulation platform
- 2 Assign DX resources for cloud deployment (Phase 2-3)
- 3 Sponsor a pilot run with one business unit (Hair or LHC)

The MVP is built. The methodology is proven.

The question is not whether to simulate — it's how fast we can scale.